Solution to Exercise 5B: Regularisation can be crucial

a

Let L be the unregularised loss according to Equation 2 in the assignment, and expand the sum over the training examples:

$$\begin{aligned} L(\mathbf{w};\mathscr{D}) &= \sum_{i=1}^{N} \exp(-y_i(\mathbf{w}\cdot\mathbf{x}_i)) \\ &= \exp 0 + \exp(w_1) + \exp(w_2) + \exp(-w_1 - w_2) \end{aligned}$$

Now

$$\begin{aligned} \frac{\partial L}{\partial w_1} &= \exp(w_1) - \exp(-w_1 - w_2) \\ \frac{\partial L}{\partial w_2} &= \exp(w_2) - \exp(-w_1 - w_2) \end{aligned}$$

To minimise the loss (or, equivalently, maximise the log-likelihood), we set the derivatives to zero and solve the resulting system of equations. Subtracting the two equations, we find that $e^{w_1} = e^{w_2}$ and therefore $w_1 = w_2$, so the weights must be equal. Substituting $w_1 = w_2 = w$ into the first equation, we find that $e^w = e^{-2w}$ and therefore $w = -2w$, implying $w = 0$. The log-likelihood has a stationary point when both weights are 0.

If we wish to establish formally that this stationary point is a minimum, we can compute the Hessian at $(0,0)$:

$$H = \begin{pmatrix} e^{w_1} + e^{-w_1-w_2} & e^{-w_1-w_2} \\ e^{-w_1-w_2} & e^{w_2} + e^{-w_1-w_2} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

where the last equality comes from substituting the values $w_1 = w_2 = 0$ we found above. Using `numpy.linalg.eig` or some other method, we find that the eigenvalues of the Hessian are $\{1,3\}$. Since both eigenvalues are positive, the Hessian is positive definite and the stationary point is a minimum.

b

Let L be the same unregularised exponential loss as above and expand the sum over the training examples with the additional feature added:

$$\begin{aligned} L(\mathbf{w};\mathscr{D}) &= \sum_{i=1}^{N} \exp(-y_i(\mathbf{w}\cdot\mathbf{x}_i)) \\ &= \exp(w_3) + \exp(w_1) + \exp(w_2) + \exp(-w_1 - w_2) \end{aligned}$$

Now $\partial L/\partial w_1$ and $\partial L/\partial w_2$ are the same as above, but

$$\frac{\partial L}{\partial w_3} = \exp(w_3) > 0.$$

Since the exponential is strictly positive, this partial derivative can never be zero, and the function has no stationary points and therefore no minima. If we try to optimise it numerically, what happens is that w_3 gets pushed towards more and more extreme negative values without ever converging.

c

Let L be the ℓ_2-regularised exponential loss:

$$\begin{aligned} L(\mathbf{w};\mathscr{D}) &= \sum_{i=1}^{N} \exp(-y_i(\mathbf{w}\cdot\mathbf{x}_i)) + ||\mathbf{w}||^2 \\ &= \exp(w_3) + \exp(w_1) + \exp(w_2) + \exp(-w_1 - w_2) \\ &\quad + w_1^2 + w_2^2 + w_3^2 \end{aligned}$$

Now

$$\begin{aligned} \frac{\partial L}{\partial w_1} &= \exp(w_1) - \exp(-w_1 - w_2) + 2w_1 \\ \frac{\partial L}{\partial w_2} &= \exp(w_2) - \exp(-w_1 - w_2) + 2w_2 \\ \frac{\partial L}{\partial w_3} &= \exp(w_3) + 2w_3 \end{aligned}$$

Setting this to zero, the first two equations are still satisfied by $(w_1, w_2) = (0,0)$. But now $\partial L/\partial w_3$ also consists of two terms that can be balanced to sum to zero by setting w_3 to a suitable negative value. The solution can be found numerically and is approximately $w_3 = -0.3517$.